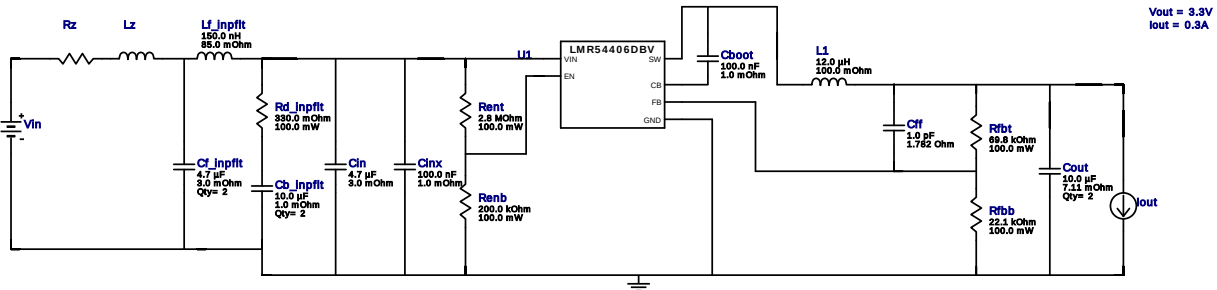


WEBENCH® Design Report

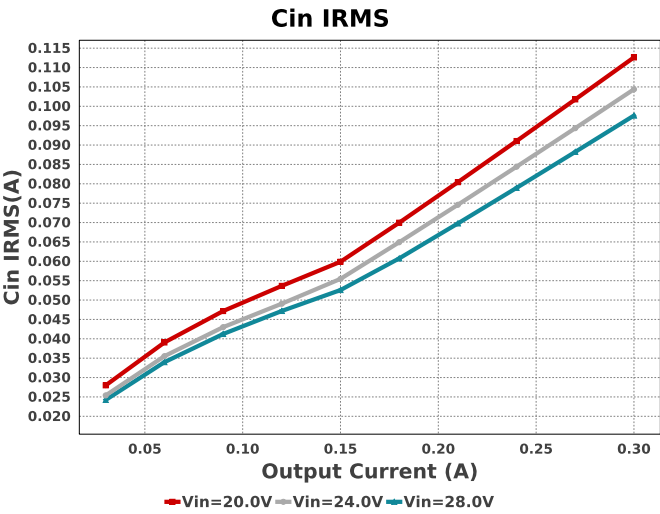
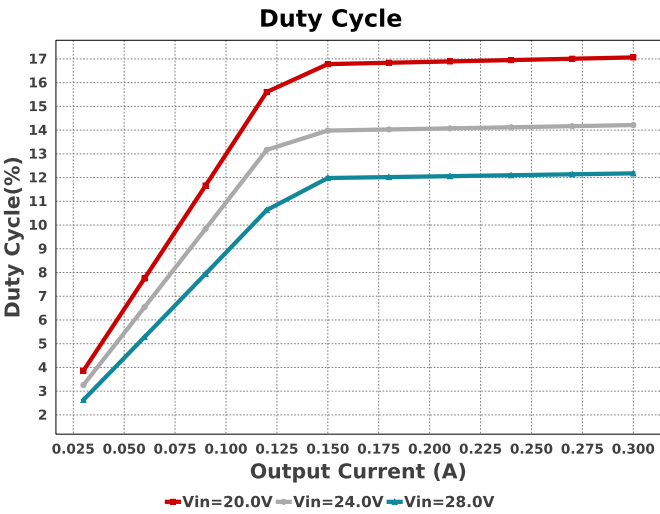
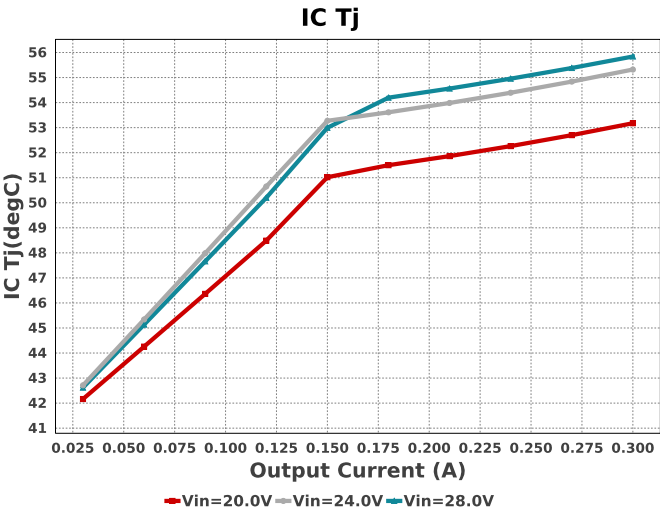
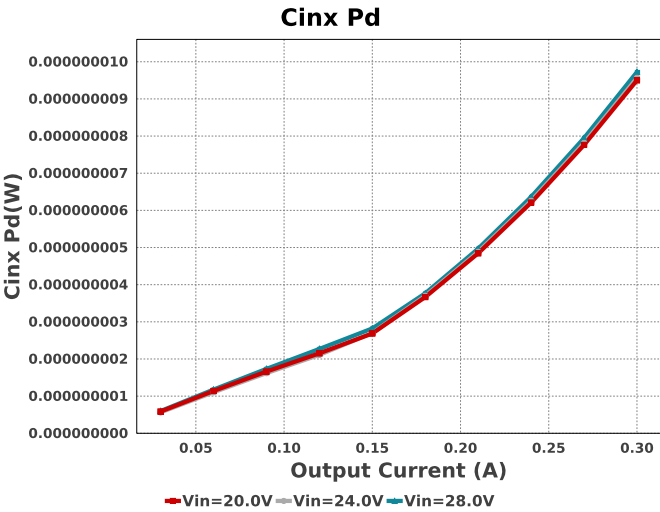
Design : 11 LMR54406DBVR
LMR54406DBVR 6V-36V to 3.30V @ 0.5A

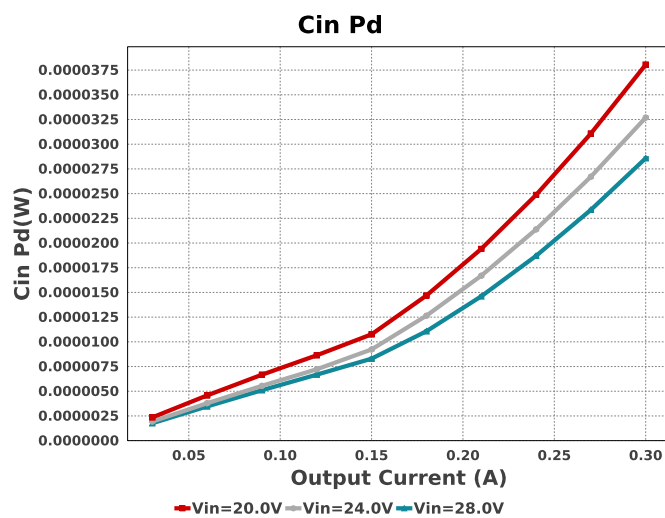
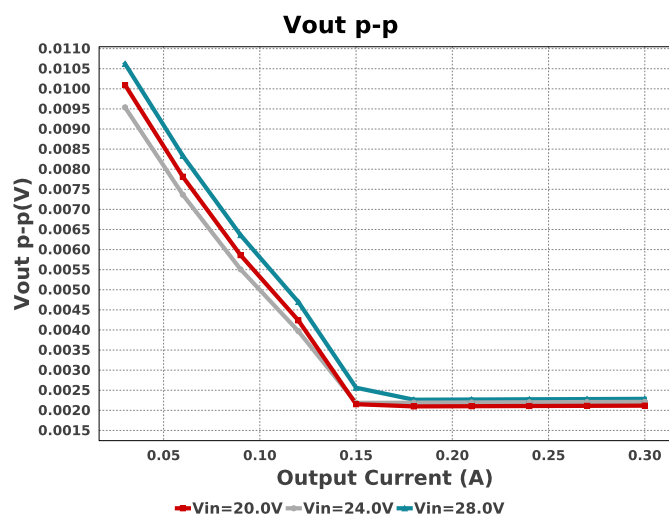
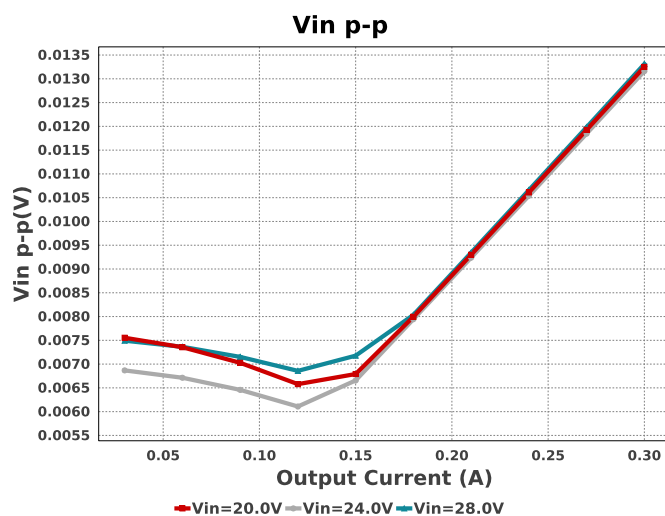
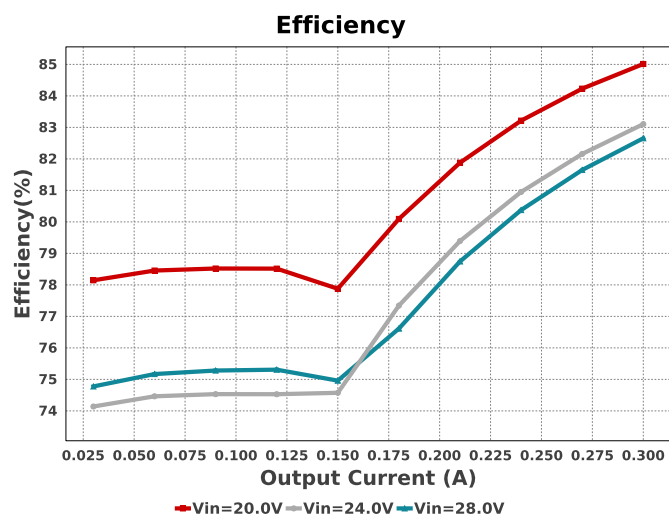
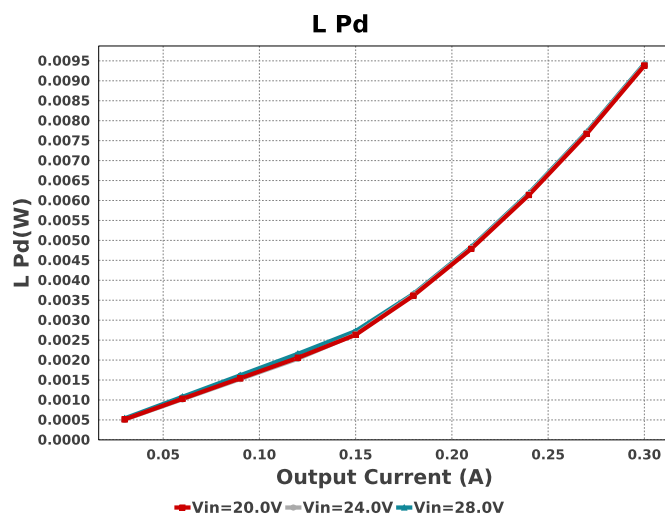
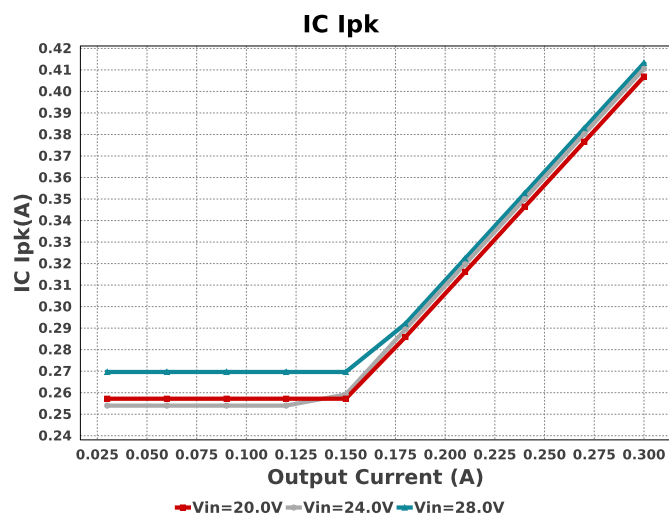


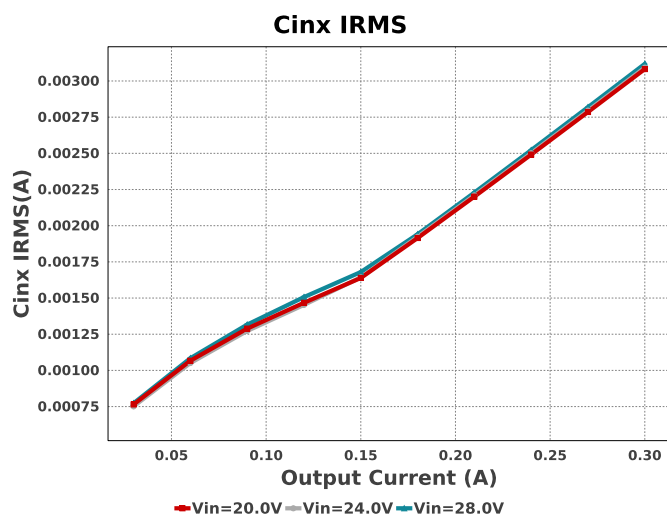
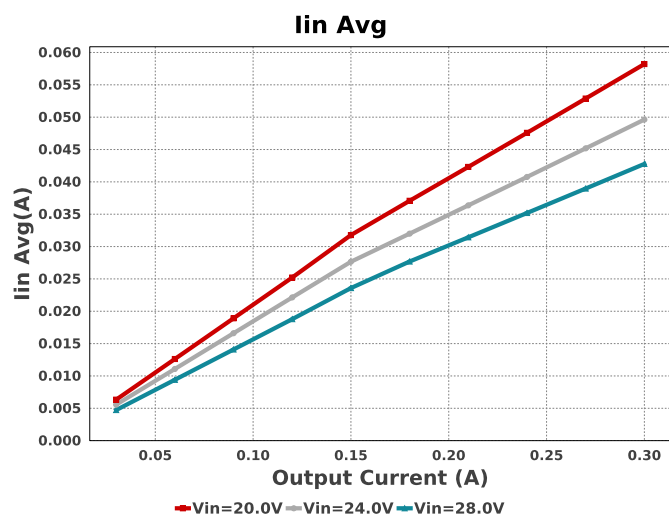
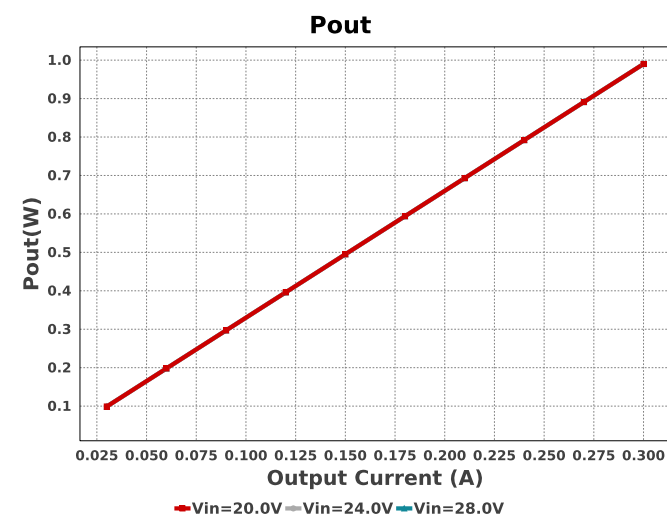
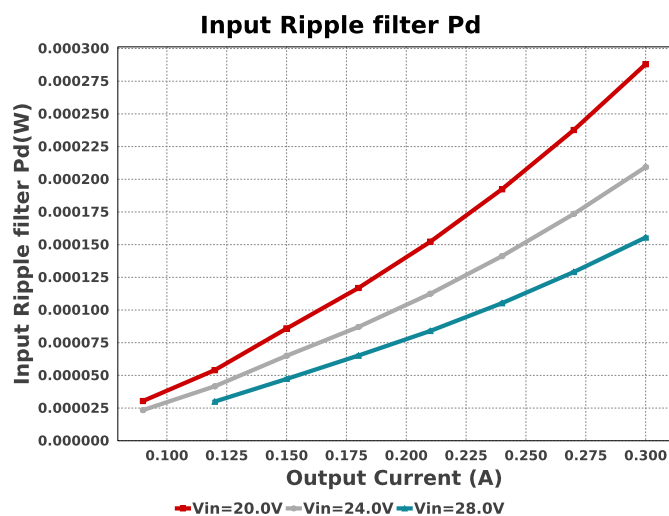
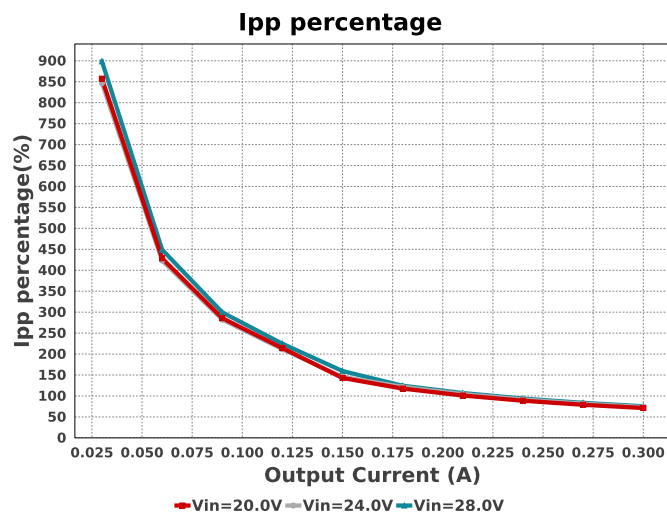
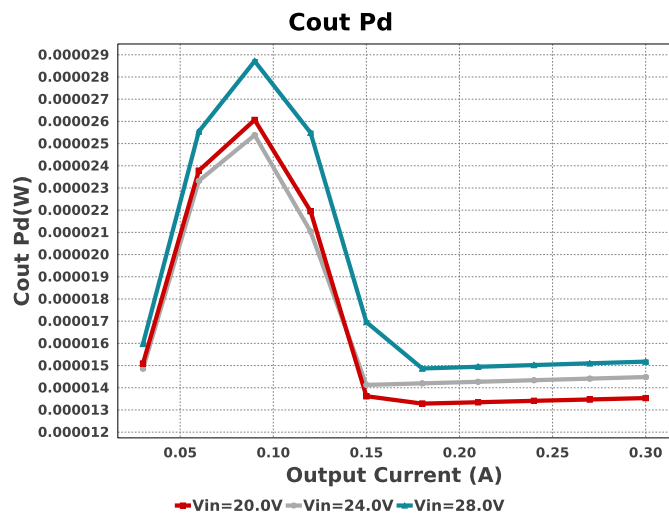
Electrical BOM

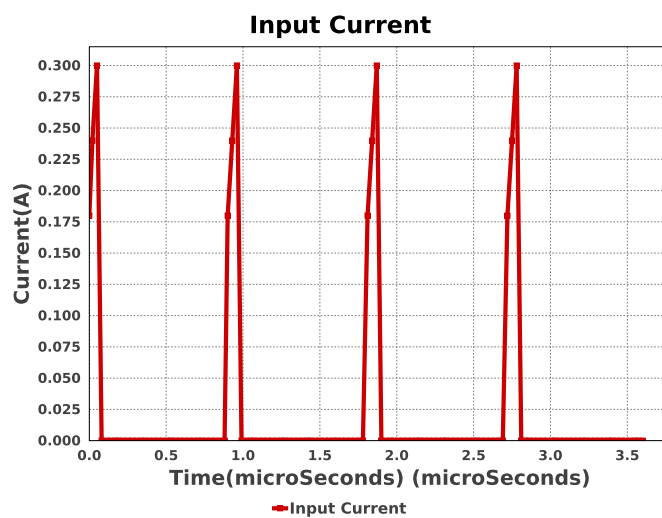
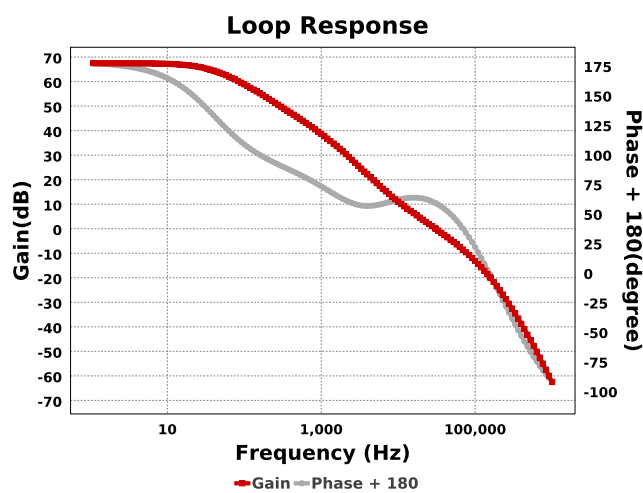
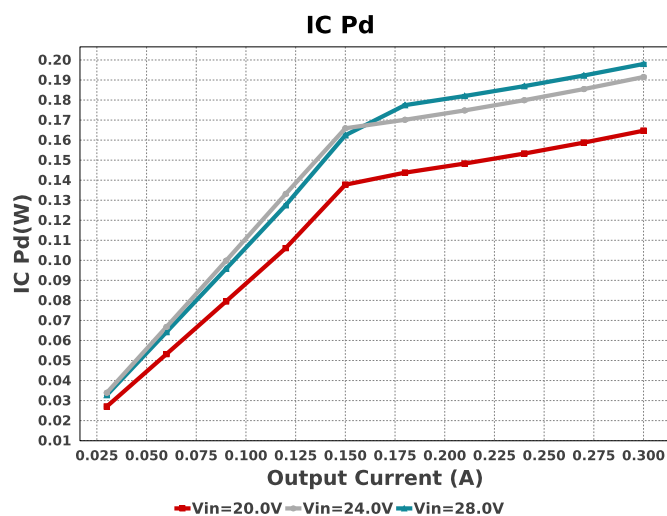
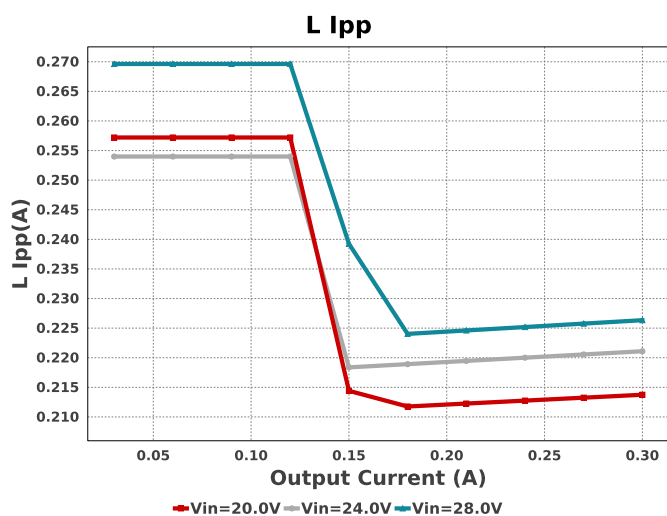
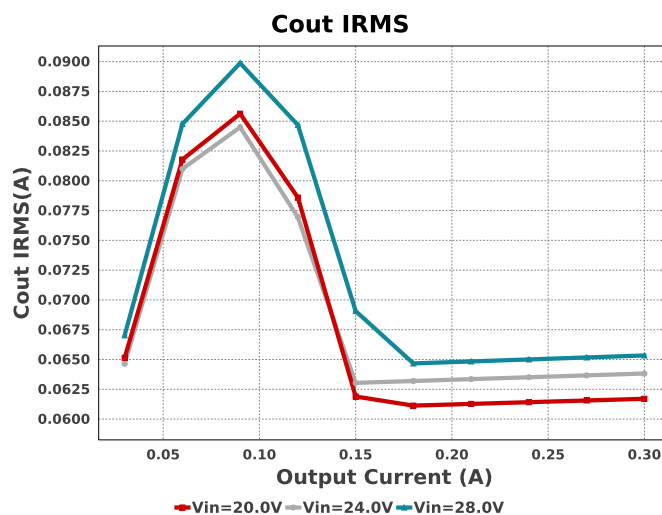
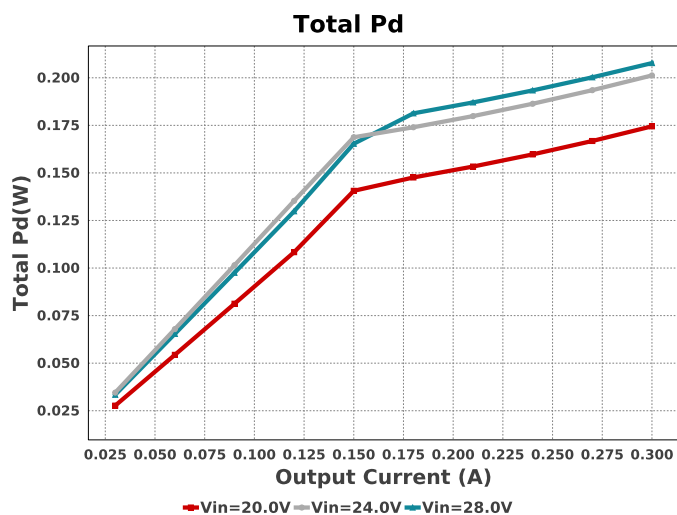
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cb_inpfilt	TDK	C3225X7R1H106M250AC Series= X7R	Cap= 10.0 uF ESR= 1.0 mOhm VDC= 50.0 V IRMS= 5.0 A	2	\$0.27	 1210 15 mm ²
Cboot	Kemet	C0603C104M4RACTU Series= X7R	Cap= 100.0 nF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	\$0.01	 0603 5 mm ²
Cf_inpfilt	MuRata	GRM31CR71H475KA12L Series= X7R	Cap= 4.7 uF ESR= 3.0 mOhm VDC= 50.0 V IRMS= 4.98 A	2	\$0.10	 1206 11 mm ²
Cff	AVX	06035A1R0CAT2A Series= C0G/NP0	Cap= 1.0 pF ESR= 1.782 Ohm VDC= 50.0 V IRMS= 0.0 A	1	\$0.01	 0603 5 mm ²
Cin	MuRata	GRM31CR71H475KA12L Series= X7R	Cap= 4.7 uF ESR= 3.0 mOhm VDC= 50.0 V IRMS= 4.98 A	1	\$0.10	 1206 11 mm ²
Cinx	MuRata	GRM188R72A104KA35D Series= X7R	Cap= 100.0 nF ESR= 1.0 mOhm VDC= 100.0 V IRMS= 3.85 A	1	\$0.04	 0603 5 mm ²
Cout	Taiyo Yuden	MSASJ219JB5106KTNA01 Series= X5R	Cap= 10.0 uF ESR= 7.11 mOhm VDC= 6.3 V IRMS= 2.7429 A	2	\$0.03	 0805 7 mm ²
L1	NIC Components	NPI54C120MTRF	L= 12.0 uH 100.0 mOhm	1	\$0.16	 IND_NPI54C 61 mm ²
Lf_inpfilt	Coilcraft	XFL2005-151MEB	L= 150.0 nH 85.0 mOhm	1	\$0.55	 XFL2005 10 mm ²
Rd_inpfilt	Panasonic	ERJ-3RQFR33V Series= ERJ-3R	Res= 330.0 mOhm Power= 100.0 mW Tolerance= 1.0%	1	\$0.02	 0603 5 mm ²
Renb	Vishay-Dale	CRCW0603200KFKEA Series= CRCW..e3	Res= 200.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	\$0.01	 0603 5 mm ²

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Rent	Vishay-Dale	CRCW06032M80FKEA Series= CRCW..e3	Res= 2.8 MOhm Power= 100.0 mW Tolerance= 1.0%	1	\$0.01	 0603 5 mm²
Rfbb	Susumu Co Ltd	RG1608P-2212-B-T5 Series= RG1608	Res= 22.1 kOhm Power= 100.0 mW Tolerance= 0.1%	1	\$0.06	 0603 5 mm²
Rfbt	Vishay-Dale	CRCW060369K8FKEA Series= CRCW..e3	Res= 69.8 kOhm Power= 100.0 mW Tolerance= 1.0%	1	\$0.01	 0603 5 mm²
U1	Texas Instruments	LMR54406DBVR	Switcher	1	\$0.47	 DBV0006A 15 mm²

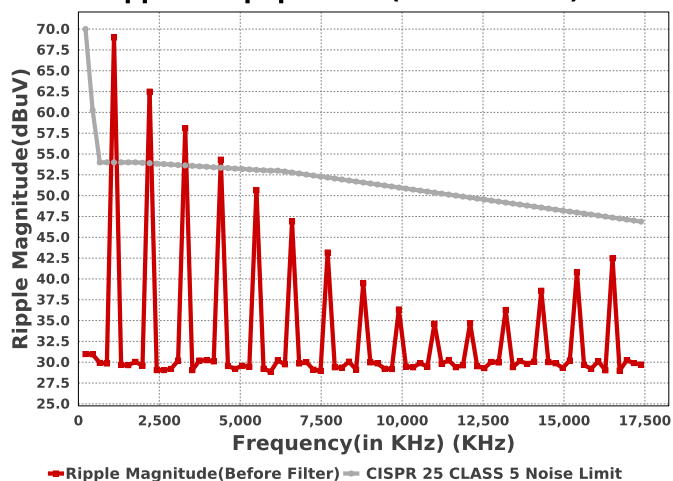




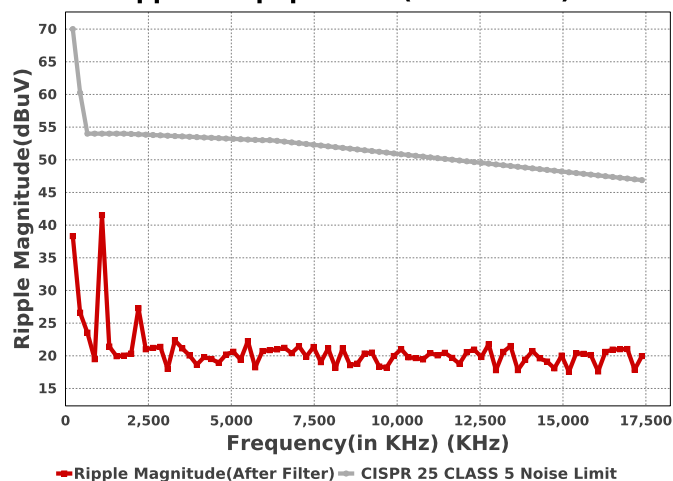




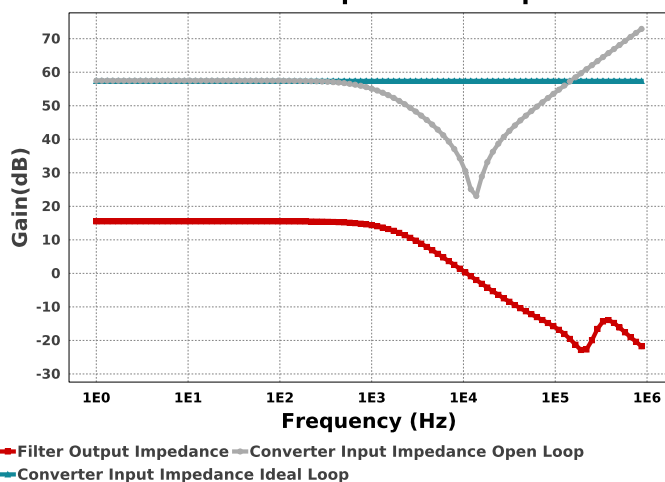
Ripple Freq Spectrum(Before Filter)



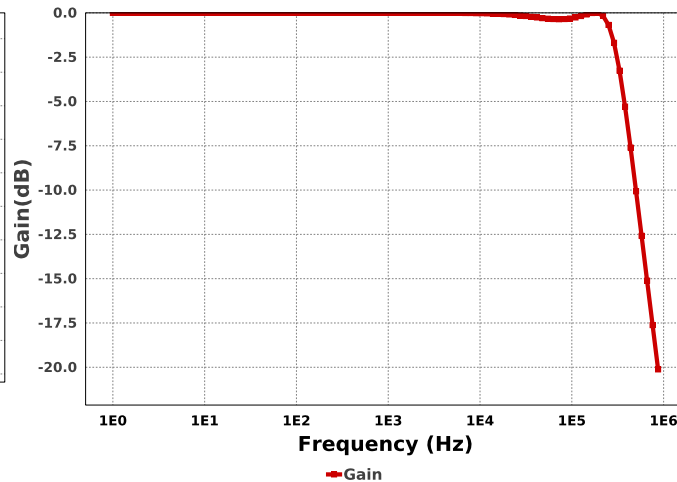
Ripple Freq Spectrum(After Filter)



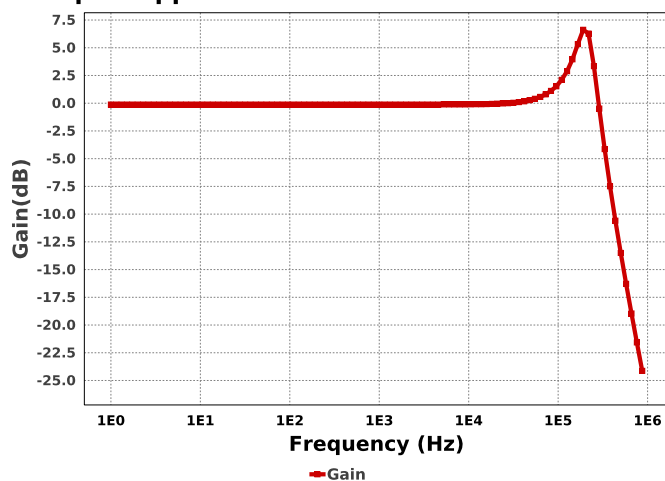
Filter vs Converter Impedance Comparison



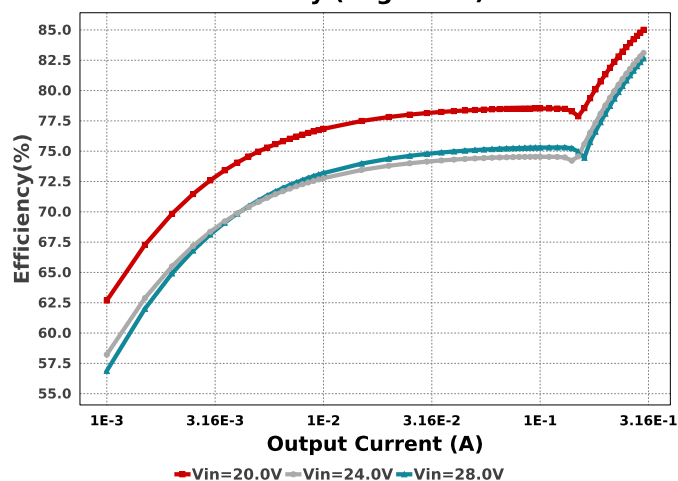
Filter Forward Transfer Function



Input Ripple Attenuation Transfer Function



Efficiency (Log-Scale)



Operating Values

#	Name	Value	Category	Description
1.	BOM Count	18		Total Design BOM count
2.	Total BOM	\$2.243		Total BOM Cost
3.	Cin IRMS	20.811 mA	Capacitor	Input capacitor RMS ripple current
4.	Cin Pd	1.299 μ W	Capacitor	Input capacitor power dissipation
5.	Cinx IRMS	615.61 μ A	Capacitor	Bulk capacitor RMS ripple current
6.	Cinx Pd	378.98 μ W	Capacitor	Bulk capacitor power dissipation
7.	Cout IRMS	54.649 mA	Capacitor	Output capacitor RMS ripple current
8.	Cout Pd	10.617 μ W	Capacitor	Output capacitor power dissipation
9.	Input Ripple Noise before input filter	29.45 dBuV	EMI Noise	Input Ripple Noise before filter at switching frequency

#	Name	Value	Category	Description
10.	Noise limits defined by CISPR Standards	70.0 dBuV	EMI Noise	Noise limits for CLASS 5 of CISPR 25 standard
11.	IC Ipk	253.992 mA	IC	Peak switch current in IC
12.	IC Pd	22.965 mW	IC	IC power dissipation
13.	IC Tj	41.837 degC	IC	IC junction temperature
14.	IC Tolerance	20.0 mV	IC	IC Feedback Tolerance
15.	ICThetaJA Effective	80.0 degC/W	IC	Effective IC Junction-to-Ambient Thermal Resistance
16.	Iin Avg	3.726 mA	IC	Average input current
17.	Ipp percentage	1.27 k%	Inductor	Inductor ripple current percentage (with respect to average inductor current)
18.	L Ipp	253.992 mA	Inductor	Peak-to-peak inductor ripple current
19.	L Pd	338.66 µW	Inductor	Inductor power dissipation
20.	Cin Pd	1.299 µW	Power	Input capacitor power dissipation
21.	Cinx Pd	378.98 pW	Power	Bulk capacitor power dissipation
22.	Cout Pd	10.617 µW	Power	Output capacitor power dissipation
23.	IC Pd	22.965 mW	Power	IC power dissipation
24.	L Pd	338.66 µW	Power	Inductor power dissipation
25.	Total Pd	23.434 mW	Power	Total Power Dissipation
26.	Duty Cycle	2.17 %	System Information	Duty cycle
27.	Efficiency	73.797 %	System Information	Steady state efficiency
28.	FootPrint	199.0 mm ²	System Information	Total Foot Print Area of BOM components
29.	Frequency	147.316 kHz	System Information	Switching frequency
30.	Iout	20.0 mA	System Information	Iout operating point
31.	Mode	PFM	System Information	Conduction Mode
32.	Pout	66.0 mW	System Information	Total output power
33.	Vin	24.0 V	System Information	Vin operating point
34.	Vin p-p	6.896 mV	System Information	Peak-to-peak input voltage
35.	Vout	3.3 V	System Information	Operational Output Voltage
36.	Vout Actual	3.327 V	System Information	Vout Actual calculated based on selected voltage divider resistors
37.	Vout Tolerance	3.357 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
38.	Vout p-p	10.335 mV	System Information	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	300.0 m	Maximum Output Current
VinMax	28.0	Maximum input voltage
VinMin	20.0	Minimum input voltage
Vout	3.3	Output Voltage
base_pn	LMR54406	Base Product Number
source	DC	Input Source Type
Ta	40.0	Ambient temperature

WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of C_{in} and C_{out} , and the inductance and DC resistance of $L1$ before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

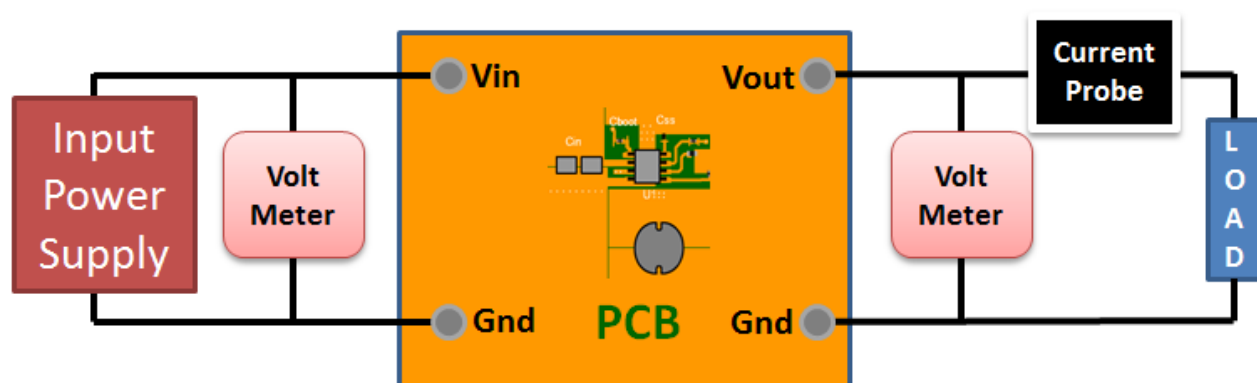
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab down to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 20.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to V_{in} and GND. Connect a digital volt meter and a load if needed to set the minimum load of the design from V_{out} and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

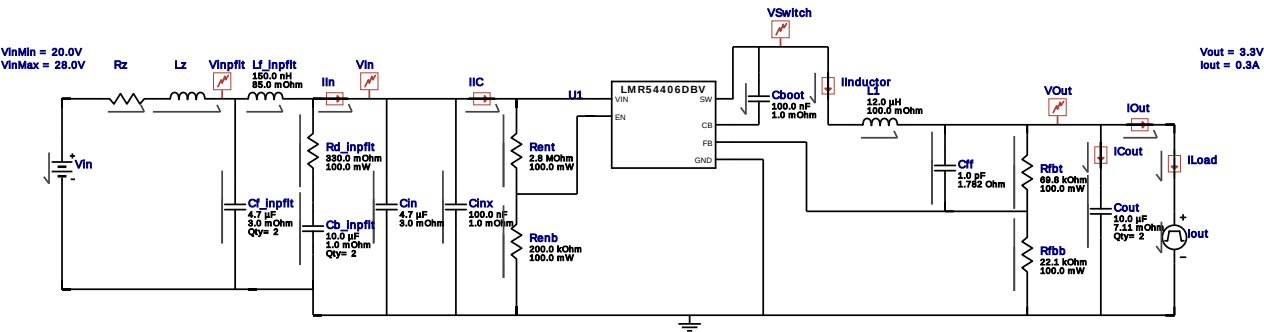
Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between V_{in} and GND, a load is connected between V_{out} and GND and a current meter is connected in series between V_{out} and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



WEBENCH® Electrical Simulation Report

Design Id = 11
sim_id = 1
Simulation Type = Load Transient



Simulation Parameters

#	Name	Parameter Name	Description	Values
1.	L1	IC	Initial Current	0.3 A
2.	Iout	signal_type	Signal Type	PULSE
		I1	Initial Load Current	0.02 A
		I2	Minimum Load Current	0.02 A
		Td	Initial Time Delay	1.0E-4 s
		Tf	Fall Time	1.0E-5 a
		Tr	Rise Time	1.0E-5 s
		Pw	Pulse Width	4.0E-4 s
3.	Rz	R	no description	5.892 Ohm
4.	Lz	L	no description	1.0E-6 H

Design Assistance

- Master key : 03B71B8129145358[v1]
- LMR54406 Product Folder : <http://www.ti.com/product/LMR54406> : contains the data sheet and other resources.

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